

Nuclear Fuel Performance

NE-533
Spring 2026

Last Time

- Solid mechanics predicts the deformation of a body from its applied load
- Elastic deformation
 - stretching of bonds
 - reversible
- Plastic deformation
 - breaking of bonds
 - permanent
 - slip and twinning

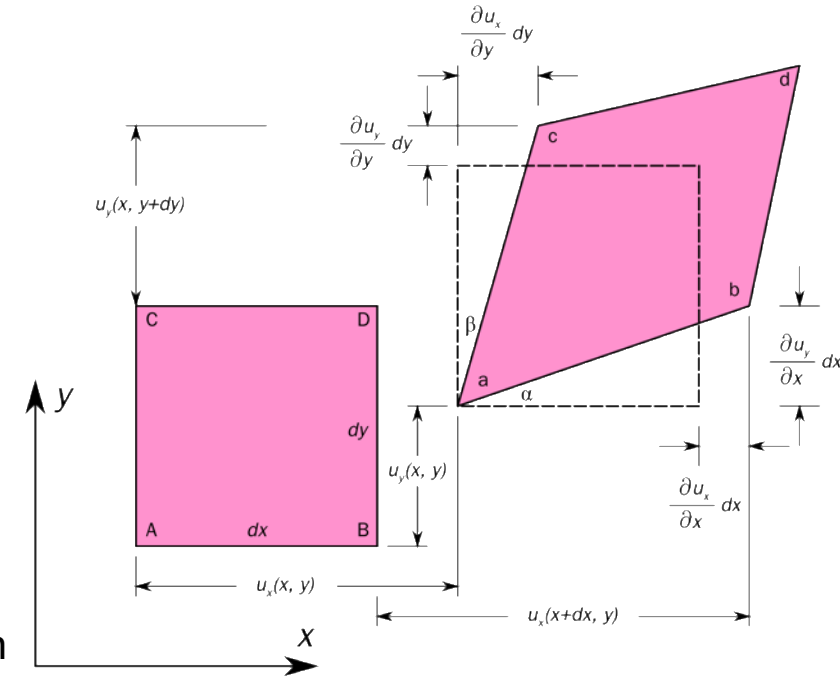
MECHANICS

Strain

- Strain is a geometrical measure of deformation representing the relative displacement between particles in a material body
- The strain in a tensile test is the deformation divided by a representative length

$$e = \frac{\Delta L}{L} = \frac{\ell - L}{L}$$

- This is engineering strain, but strain can also be defined as “true strain,” accounts for shrinking of section area and the effect of developed elongation on further elongation



- Images shows strain as a second order tensor
 - Let $\mathbf{u}$ be a vector of the displacements
 - The small strain tensor is

$$\epsilon_{ij} \equiv \frac{1}{2} \left(\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} \right)$$

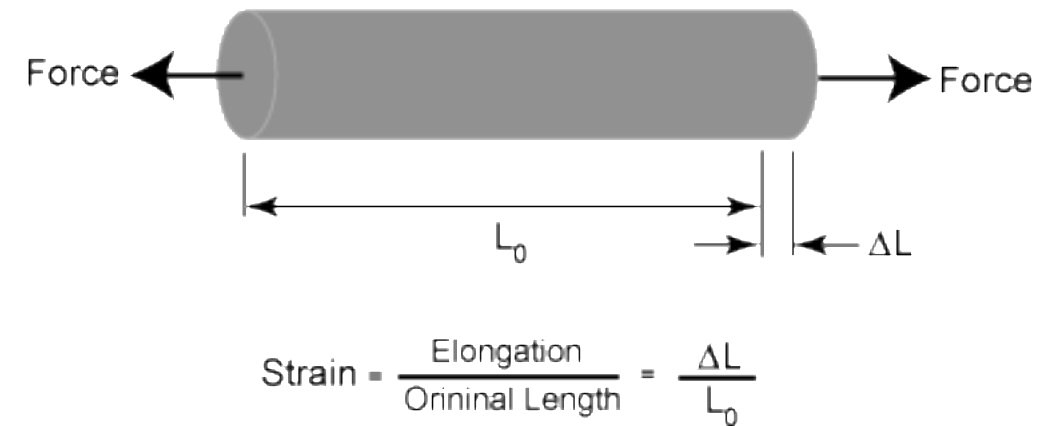
$$\begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{bmatrix} = \begin{bmatrix} \frac{\partial u_1}{\partial x_1} & \frac{1}{2} \left(\frac{\partial u_1}{\partial x_2} + \frac{\partial u_2}{\partial x_1} \right) & \frac{1}{2} \left(\frac{\partial u_1}{\partial x_3} + \frac{\partial u_3}{\partial x_1} \right) \\ \frac{1}{2} \left(\frac{\partial u_2}{\partial x_1} + \frac{\partial u_1}{\partial x_2} \right) & \frac{\partial u_2}{\partial x_2} & \frac{1}{2} \left(\frac{\partial u_2}{\partial x_3} + \frac{\partial u_3}{\partial x_2} \right) \\ \frac{1}{2} \left(\frac{\partial u_3}{\partial x_1} + \frac{\partial u_1}{\partial x_3} \right) & \frac{1}{2} \left(\frac{\partial u_3}{\partial x_2} + \frac{\partial u_2}{\partial x_3} \right) & \frac{\partial u_3}{\partial x_3} \end{bmatrix}$$

- The common strain states are the same as the stress (uniaxial tension, etc.)

Strain produces stress

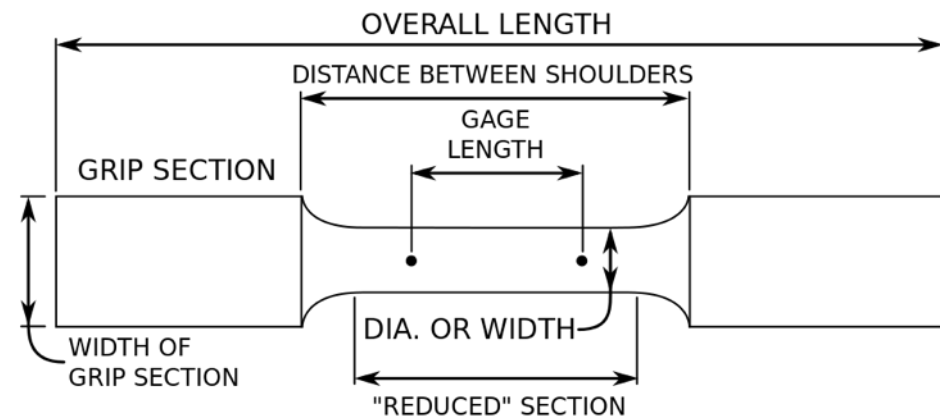
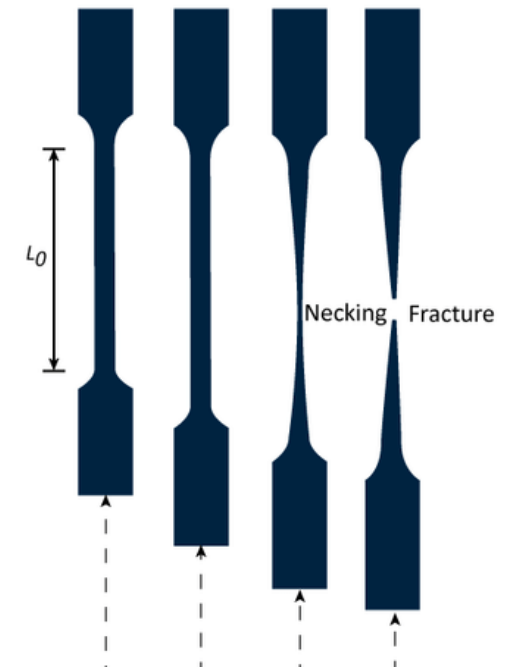
- A deformation (strain) results in stress within a material
- $\sigma = F(\epsilon)$
- For small strains, the stress is elastic and is a linear function of the strain
 - $\sigma = \mathcal{C}(\epsilon)$
- For larger deformation, some of the strain is elastic and increases the stress. The rest is plastic and does NOT contribute to the stress.
 - $\epsilon = \epsilon_e + \epsilon_p$
 - $\sigma = \mathcal{C}\epsilon_e$
- The elastic energy density in a material is a scalar quantity equal to

$$E_{el} = \frac{1}{2} \epsilon_e \cdot \sigma$$



Tensile testing

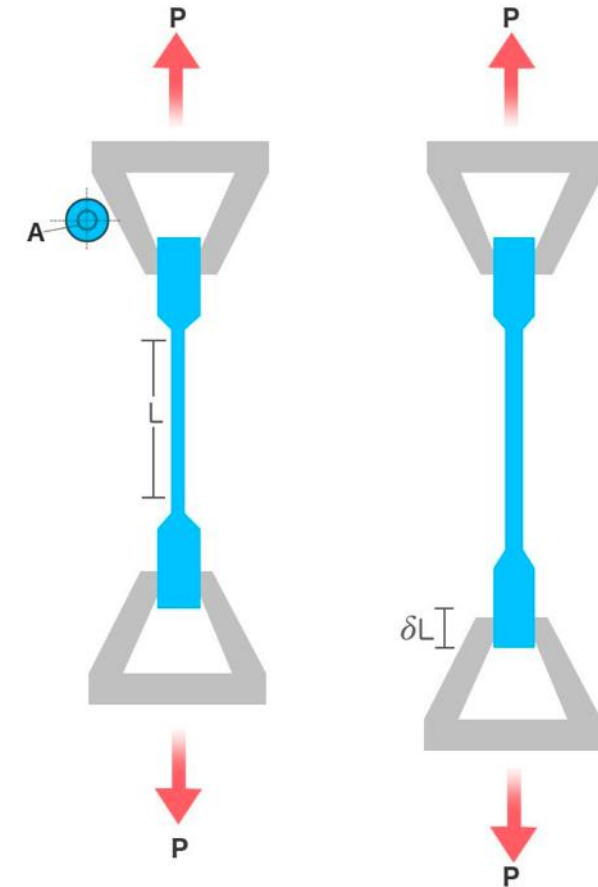
- The most common means of determining mechanical properties is a uniaxial tension test
- Apply a uniaxial load until failure
- Properties that are directly measured via a tensile test include ultimate tensile strength, maximum elongation, Young's modulus, Poisson's ratio, etc.



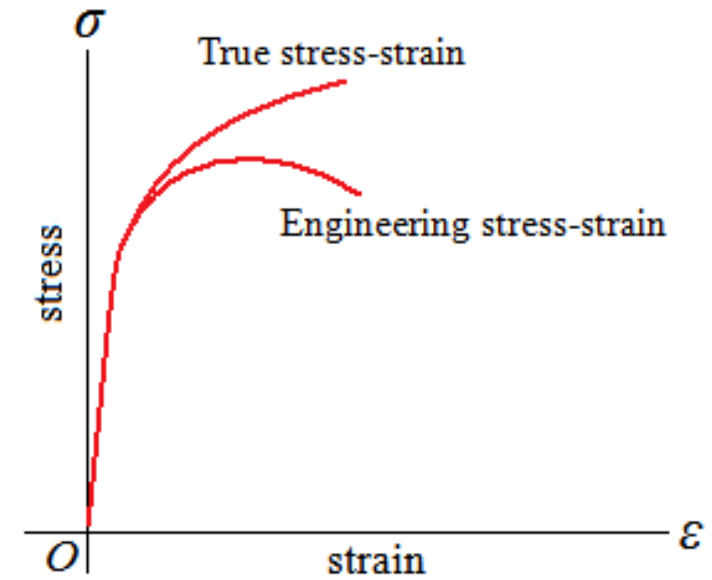
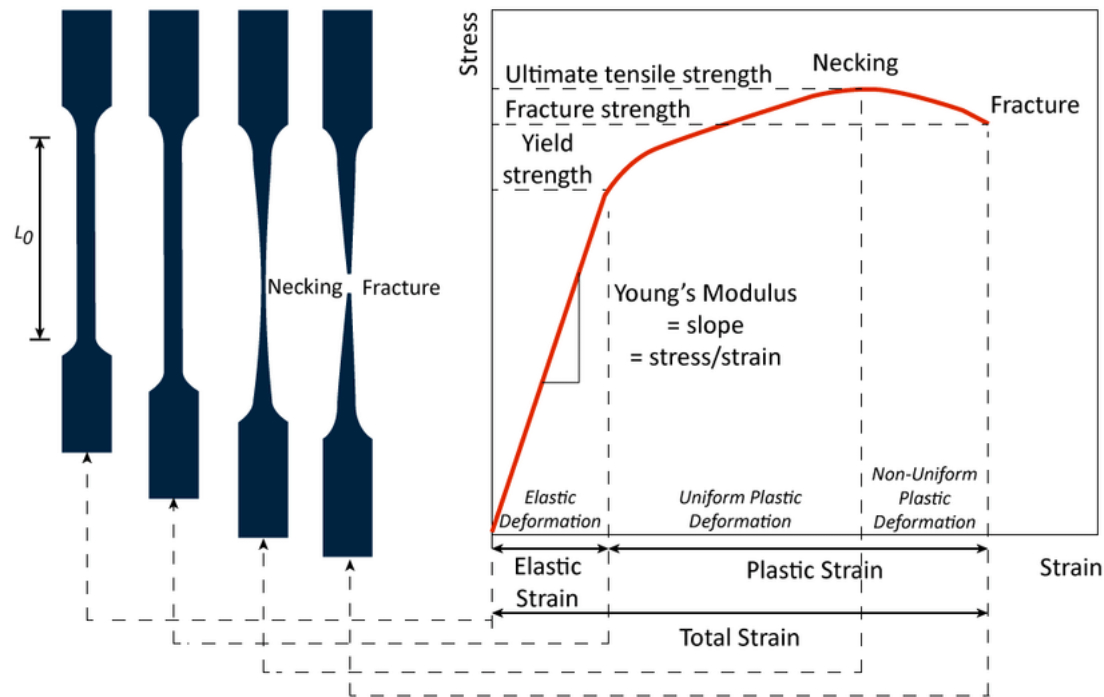
Stress/strain from a tensile test

- A_0 = Initial cross section area
- A = deformed cross section area
- P or F applied load/force

	Engineering	True
stress σ	$\frac{F}{A_0}$	$\frac{F}{A}$
strain ϵ	$\frac{l - l_0}{l_0}$	$\int_{l_0}^l \frac{dl}{l} = \ln\left(\frac{l}{l_0}\right)$



Stress vs strain curves



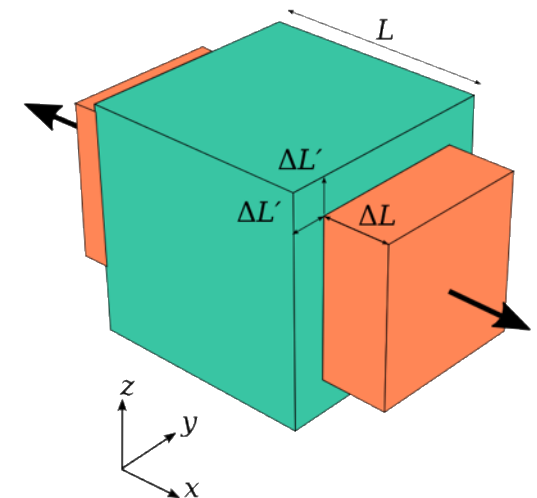
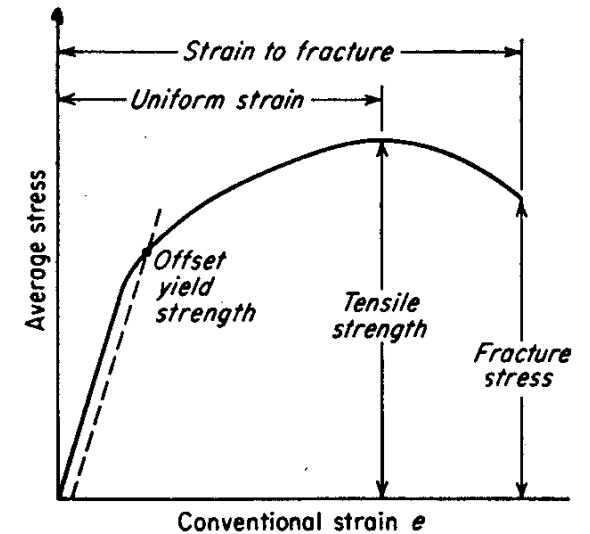
Using stress/strain curves

- In the elastic portion of the stress-strain curve, the stress varies linearly with strain
- The slope of the line is Young's Modulus, E : $\sigma = E \varepsilon$
- Yield strength is usually taken at 0.2% offset
- Poisson's ratio, ν , is the ratio of the shrinkage in cross section due to the extension in the pulling direction

$$\nu = -\frac{d\varepsilon_{\text{trans}}}{d\varepsilon_{\text{axial}}} = -\frac{d\varepsilon_y}{d\varepsilon_x} = -\frac{d\varepsilon_z}{d\varepsilon_x} \quad \nu \approx \frac{\Delta L'}{\Delta L}$$

- Can obtain the shear modulus (G) from the elastic modulus and Poisson's ratio

$$G = \frac{E}{2(1 + \nu)}$$



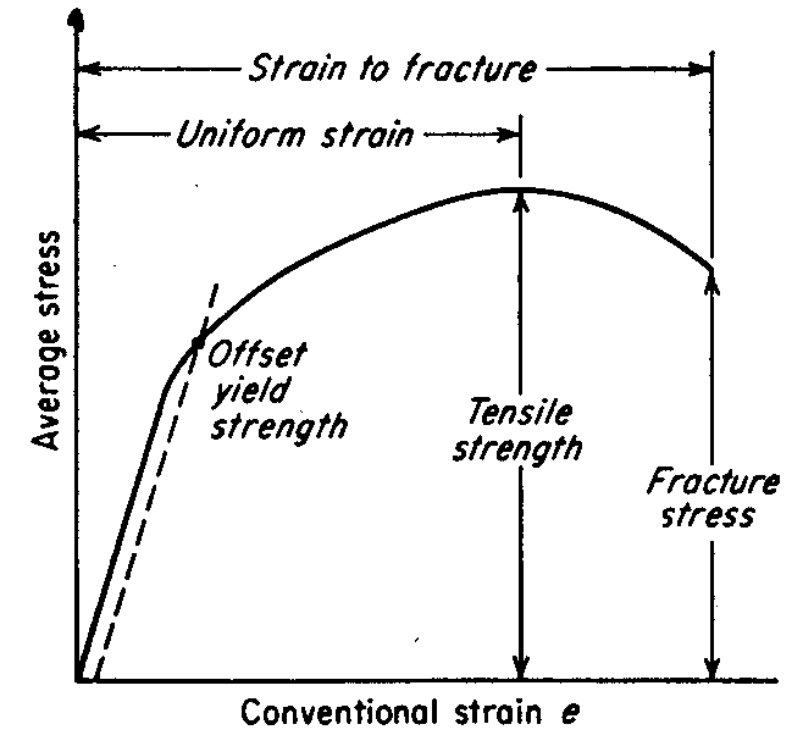
Using stress/strain curves

- The shear modulus, G , defines the stress to strain ratio in shear

$$\sigma_{12} = G\epsilon_{12}$$

- For isotropic materials, $G = E / (2(1 + \nu))$
- In matrix form, the elasticity/stiffness tensor from Hooke's law for isotropic materials can be written as

$$\begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ 2\epsilon_{23} \\ 2\epsilon_{13} \\ 2\epsilon_{12} \end{bmatrix} = \begin{bmatrix} \epsilon_{11} \\ \epsilon_{22} \\ \epsilon_{33} \\ \gamma_{23} \\ \gamma_{13} \\ \gamma_{12} \end{bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & -\nu & 0 & 0 & 0 \\ -\nu & 1 & -\nu & 0 & 0 & 0 \\ -\nu & -\nu & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2+2\nu & 0 & 0 \\ 0 & 0 & 0 & 0 & 2+2\nu & 0 \\ 0 & 0 & 0 & 0 & 0 & 2+2\nu \end{bmatrix} \begin{bmatrix} \sigma_{11} \\ \sigma_{22} \\ \sigma_{33} \\ \sigma_{23} \\ \sigma_{13} \\ \sigma_{12} \end{bmatrix}$$



$$\epsilon_{ij} = \frac{1}{E} (\sigma_{ij} - \nu(\sigma_{kk}\delta_{ij} - \sigma_{ij}))$$

$$\delta_{ij} = \begin{cases} 0 & \text{if } i \neq j, \\ 1 & \text{if } i = j. \end{cases}$$

Isotropic and Anisotropic

- Being a linear mapping between the nine numbers σ_{ij} and the nine numbers ϵ_{kl} , the stiffness tensor $\mathbf{c}$ is represented by a matrix of $3 \times 3 \times 3 \times 3 = 81$ real numbers
- Given minor symmetries of the stiffness tensor, $c_{ijkl} = c_{jikl}$, this 81 elastic constants can be reduced to 36
- Major symmetries, $c_{ijkl} = c_{klij}$, reduce this number from 36 to 21
- Isotropic materials deform the same way no matter in what direction you deform them.
 - They have 2 unique elastic constants, C_{11} and C_{12}
- Anisotropic materials behave differently in different directions
 - The elasticity tensor can have 21 unique components defining anisotropy
 - Orthotropic materials have 9 unique elastic constants
 - Cubic structured materials have 3 unique elastic constants (UO_2)
 - Hexagonal structured materials have 5 unique elastic constants
- Polycrystalline anisotropic materials can behave as isotropic, because the various grains average out

$$\boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{bmatrix}; \quad \boldsymbol{\sigma} = \begin{bmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{bmatrix}$$

$$\sigma_{ij} = - \sum_{k=1}^3 \sum_{l=1}^3 c_{ijkl} \epsilon_{kl}$$

$$\begin{bmatrix} c_{1111} & c_{1122} & c_{1133} & c_{1123} & c_{1131} & c_{1112} \\ c_{2211} & c_{2222} & c_{2233} & c_{2223} & c_{2231} & c_{2212} \\ c_{3311} & c_{3322} & c_{3333} & c_{3323} & c_{3331} & c_{3312} \\ c_{2311} & c_{2322} & c_{2333} & c_{2323} & c_{2331} & c_{2312} \\ c_{3111} & c_{3122} & c_{3133} & c_{3123} & c_{3131} & c_{3112} \\ c_{1211} & c_{1222} & c_{1233} & c_{1223} & c_{1231} & c_{1212} \end{bmatrix}$$

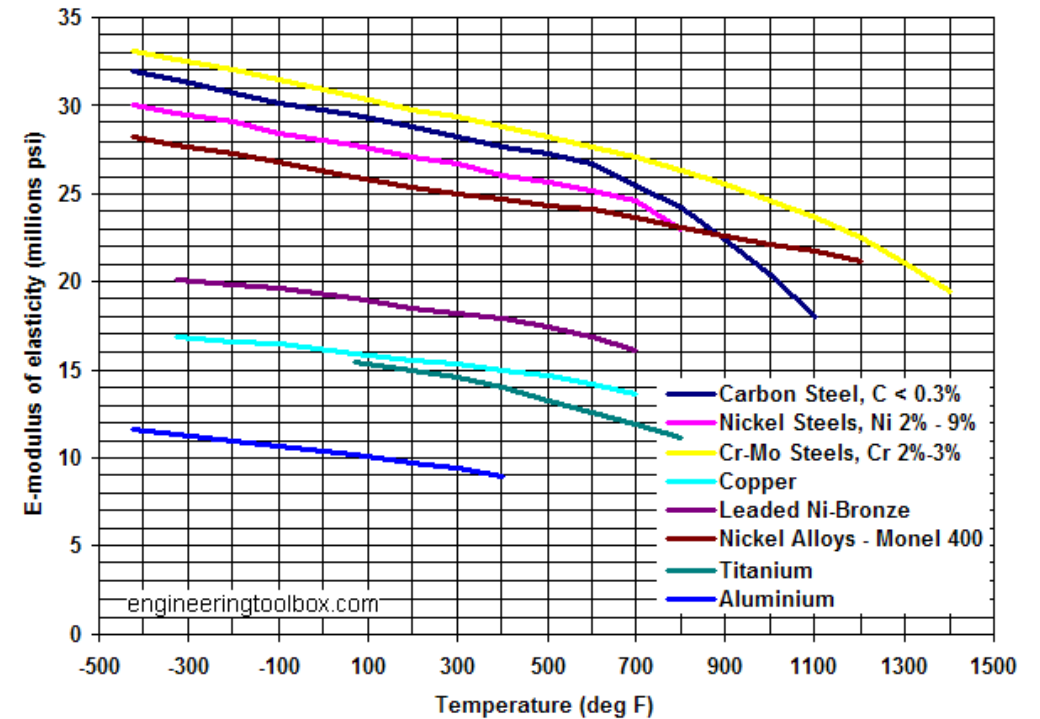
$$\begin{bmatrix} C_{11} & C_{12} & C_{13} & C_{14} & C_{15} & C_{16} \\ & C_{22} & C_{23} & C_{24} & C_{25} & C_{26} \\ & & C_{33} & C_{34} & C_{35} & C_{36} \\ & & & C_{44} & C_{45} & C_{46} \\ & & & & C_{55} & C_{56} \\ & & & & & C_{66} \end{bmatrix}$$

Isotropic elastic properties for some materials

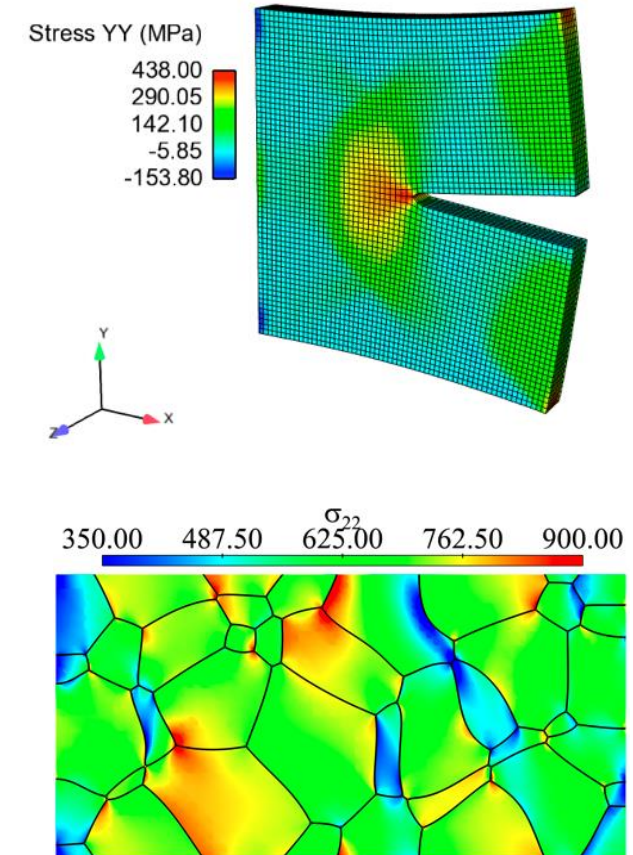
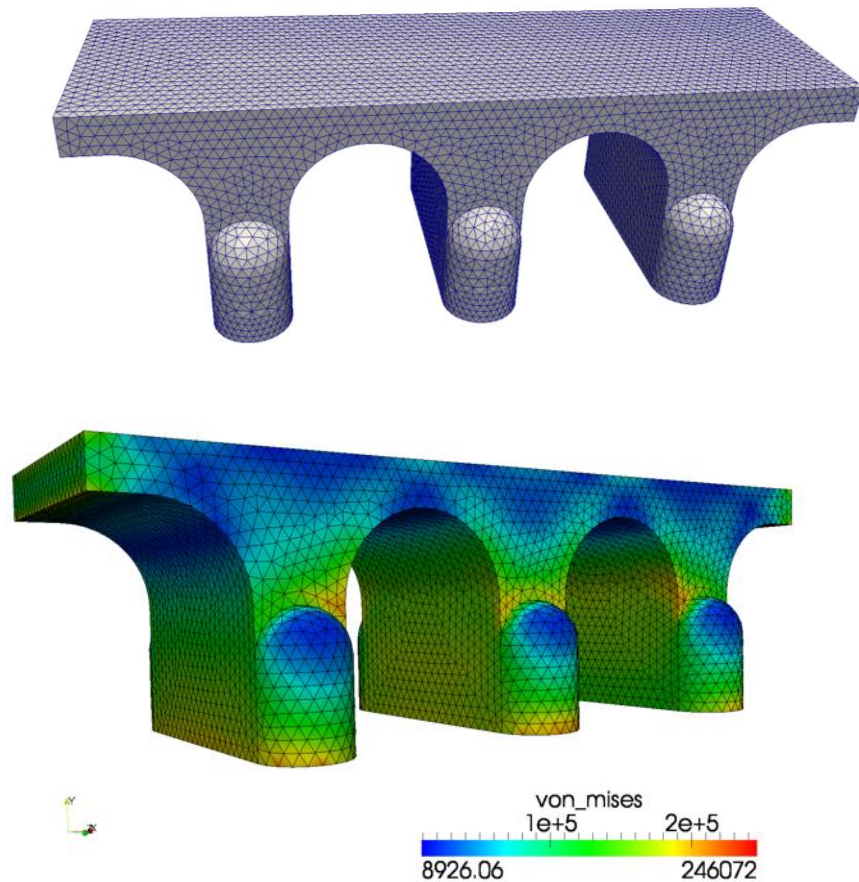
Material	E (GPa)	ν
Aluminum	70.3	0.345
Gold	78.0	0.44
Iron	211.4	0.293
Nickel	199.5	0.312
Tungsten	411.0	0.28
Zircaloy	80.0	0.41
UO ₂	200.0	0.345

Elastic constants are not “constant”

- Properties change with temperature
- The decrease in elastic constants with temperature is called softening
- Young's modulus is typically a function of temperature, decreasing with increasing temperature
- Shear Modulus and Poisson's ratio can also change with T



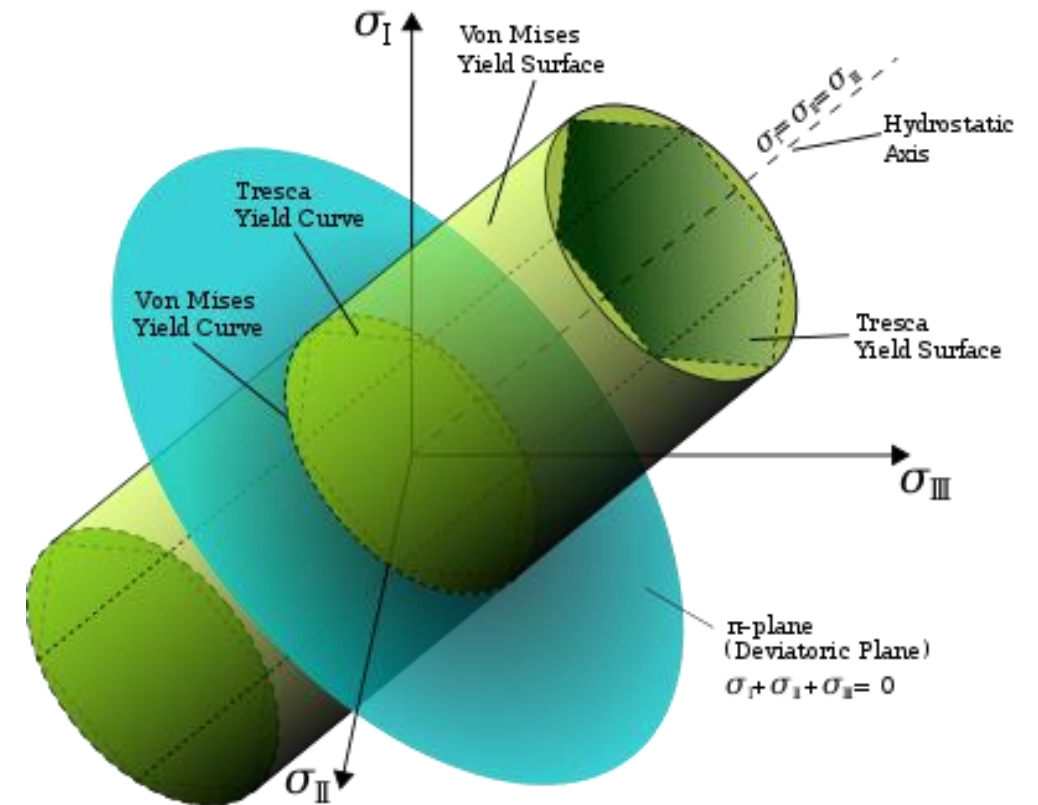
In actual materials, the stress and the strain change throughout the material



von Mises yield criterion

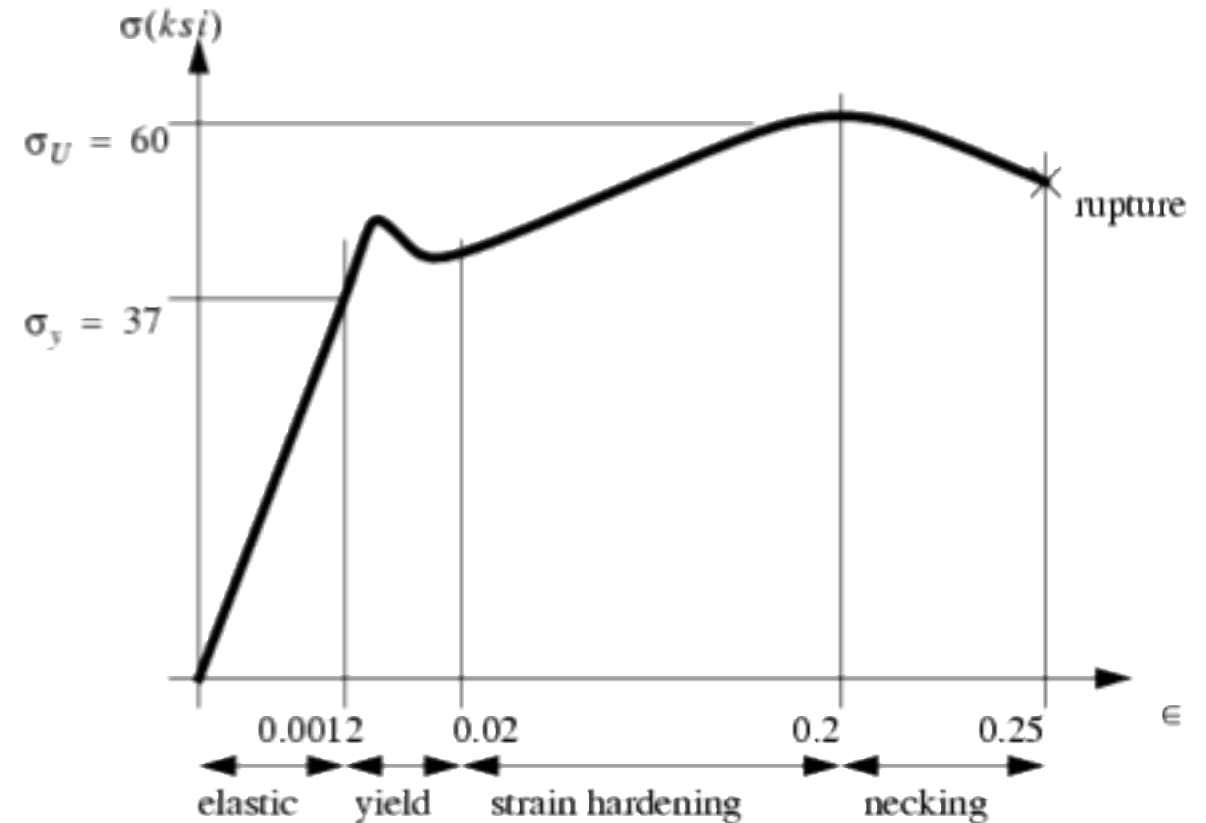
- The maximum distortion criterion (also von Mises yield criterion) states that yielding of a ductile material begins when the second invariant of deviatoric stress (J2) reaches a critical value
- A material is said to start yielding when the von Mises stress reaches the yield strength
- The von Mises stress satisfies the property where two stress states with equal distortion energy have an equal von Mises stress
- The von Mises stress is used to predict yielding of materials under complex loading and the results of uniaxial tensile tests

$$\sigma_v^2 = \frac{1}{2} [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2 + 6(\sigma_{23}^2 + \sigma_{31}^2 + \sigma_{12}^2)]$$



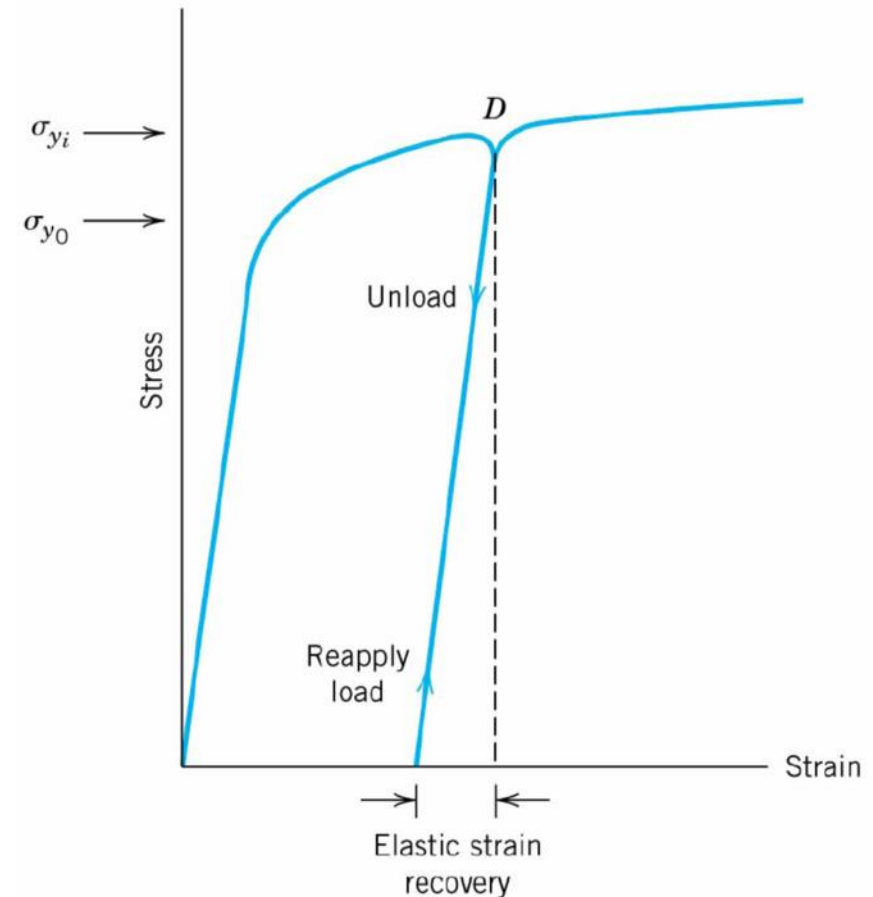
Stress/Strain Regions

- Once the stress reaches the yield stress, it plastically deforms
- σ_y is the yield stress
- σ_U is the ultimate tensile stress
- The final stress before rupture is called the fracture stress



Permanent Strain and Strain Hardening

- After plastic deformation, if you unload the sample, it still has the permanent strain
- The plastic behavior has modified the microstructure, and has often increased the yield point
- After unloading, the strain hardening has changed the material
- Thus, if you reload, the yield stress is often higher than it was previously

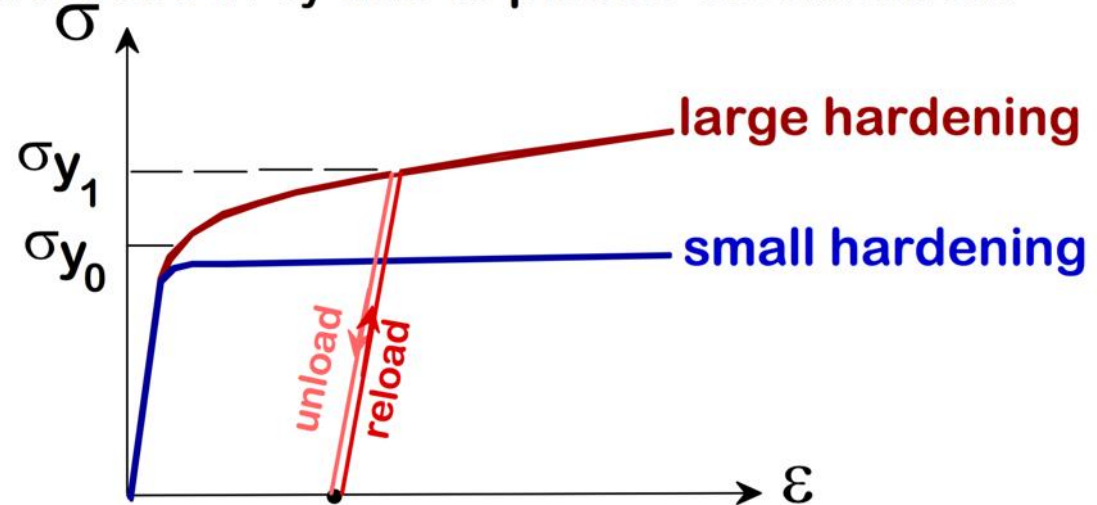


The hardening behavior changes for different materials

$$\sigma = \sigma_y + K(\epsilon_0 + \epsilon_p)^n$$

- K is a strength coefficient, n is the strain hardening exponent, ϵ_0 is the prior plastic strain, ϵ_p is the plastic strain, and σ_y is the original yield strength
- The strain hardening exponent is a material property, with a value between 0 and 1
- A value of 0 means that a material is a perfectly plastic solid, while a value of 1 represents a 100% elastic solid.

- An increase in σ_y due to plastic deformation.



- Curve fit to the stress-strain response:

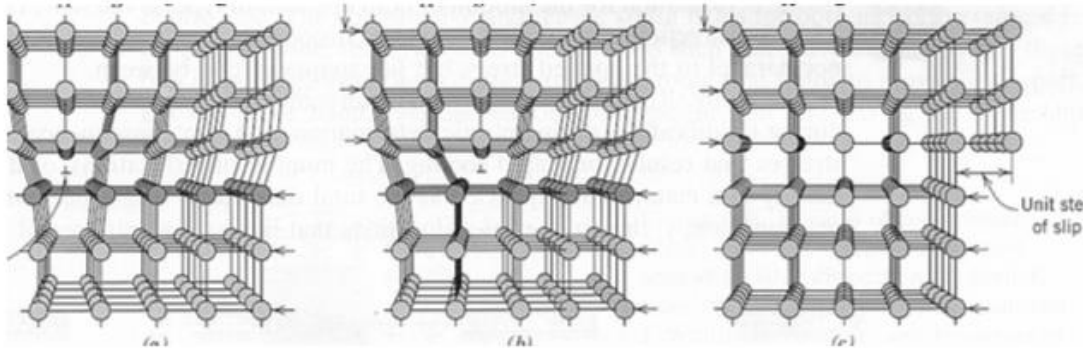
$$\sigma_T = C(\epsilon_T)^n$$

“true” stress (F/A) “true” strain: $\ln(L/L_0)$

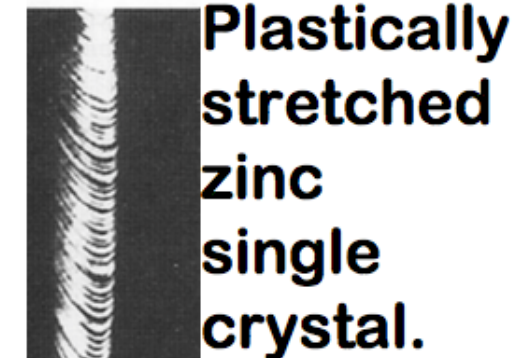
hardening exponent:
 $n=0.15$ (some steels)
 to $n=0.5$ (some copper)

Dislocation motion

- Plastic deformation occurs due to dislocation motion
- A dislocation is a line defect
 - Edge and screw type
- When it moves, only a small number of bonds are broken at a time



Adapted from Fig. 7.1, *Callister 6e*. (Fig. 7.1 is adapted from A.G. Guy, *Essentials of Materials Science*, McGraw-Hill Book Company, New York, 1976. p. 153.)



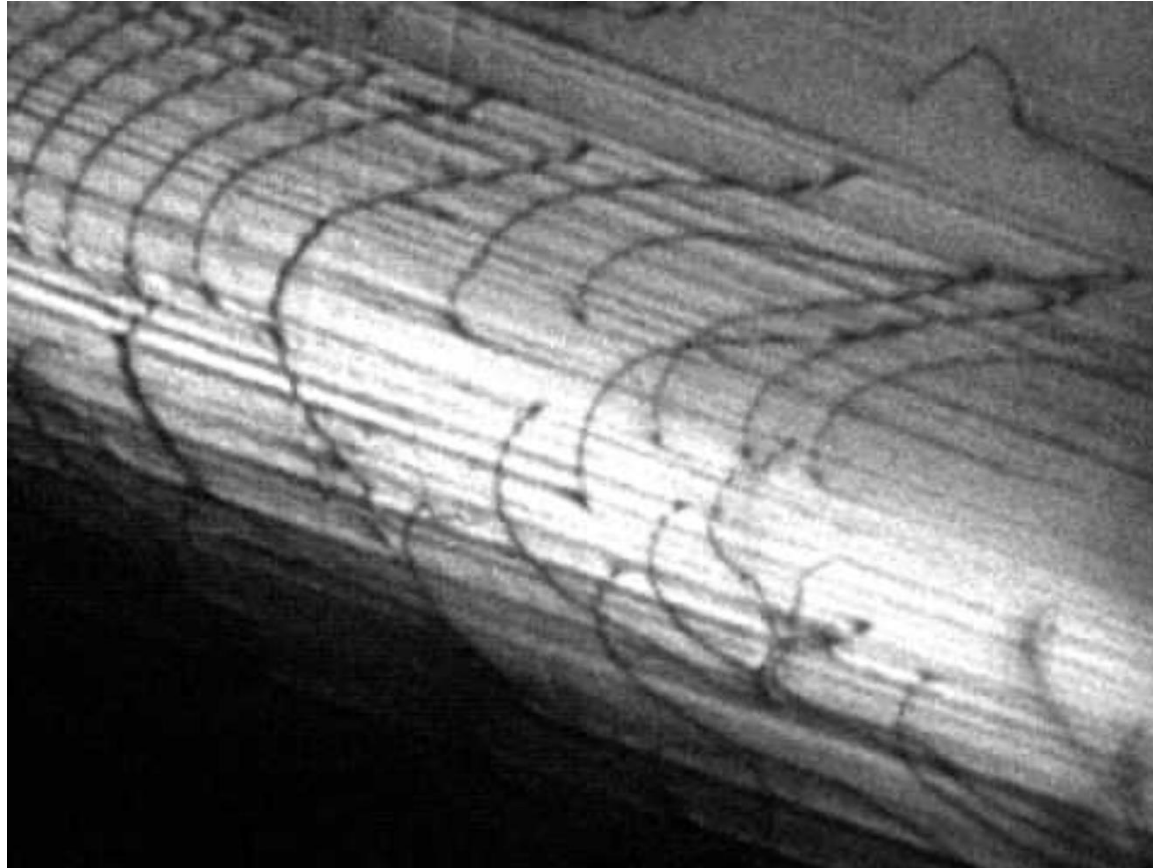
Plastically stretched zinc single crystal.

Adapted from Fig. 7.9, *Callister 6e*. (Fig. 7.9 is from C.F. Elam, *The Distortion of Metal Crystals*, Oxford University Press, London, 1935.)



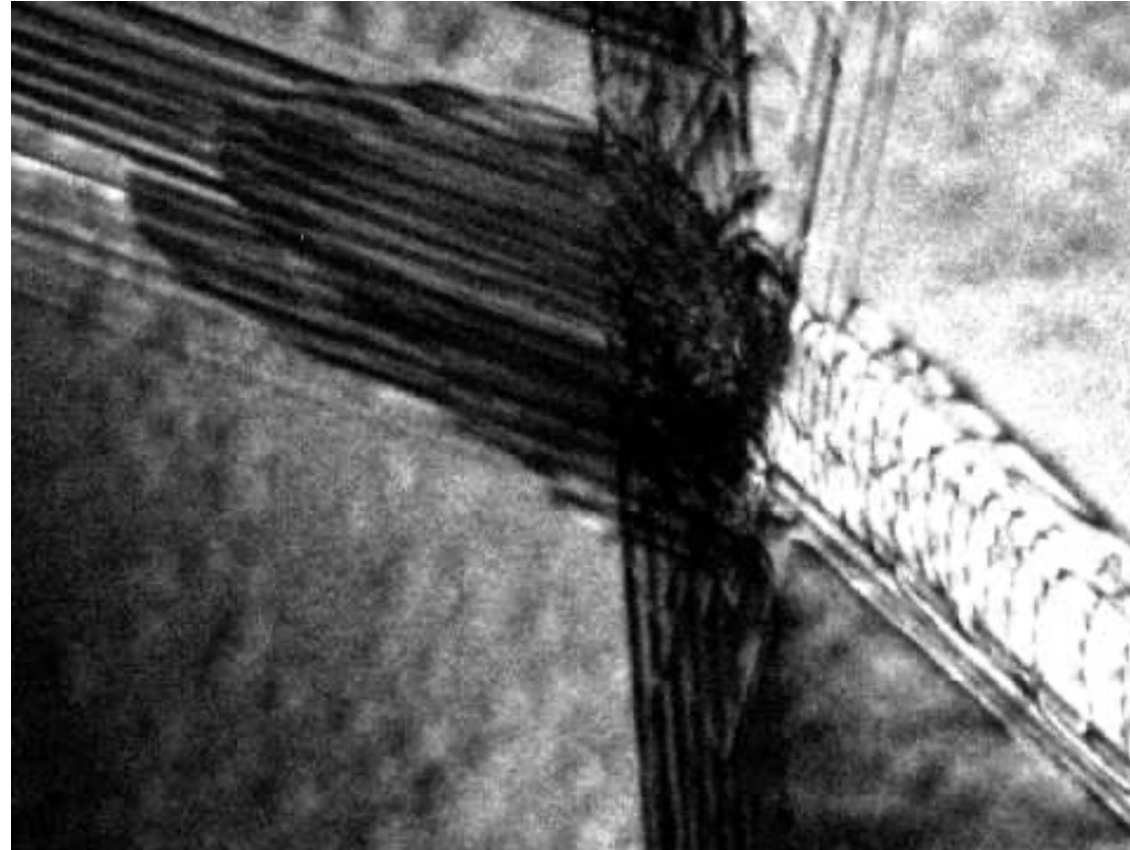
Adapted from Fig. 7.8, *Callister 6e*.

Dislocation are produced and move during deformation



https://www.youtube.com/watch?v=EXbiEopDJ_g

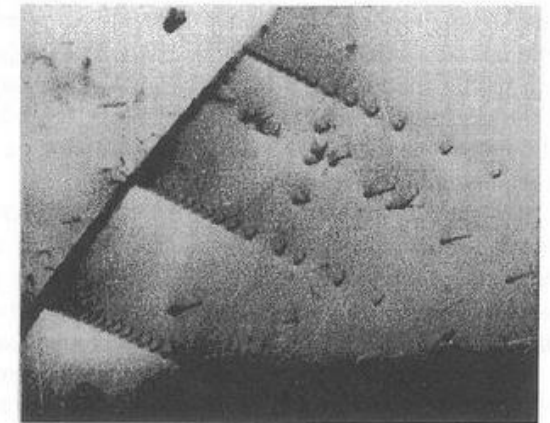
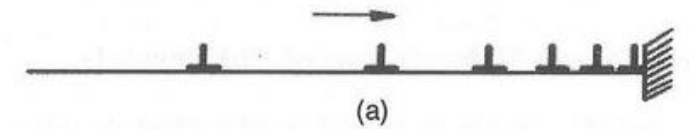
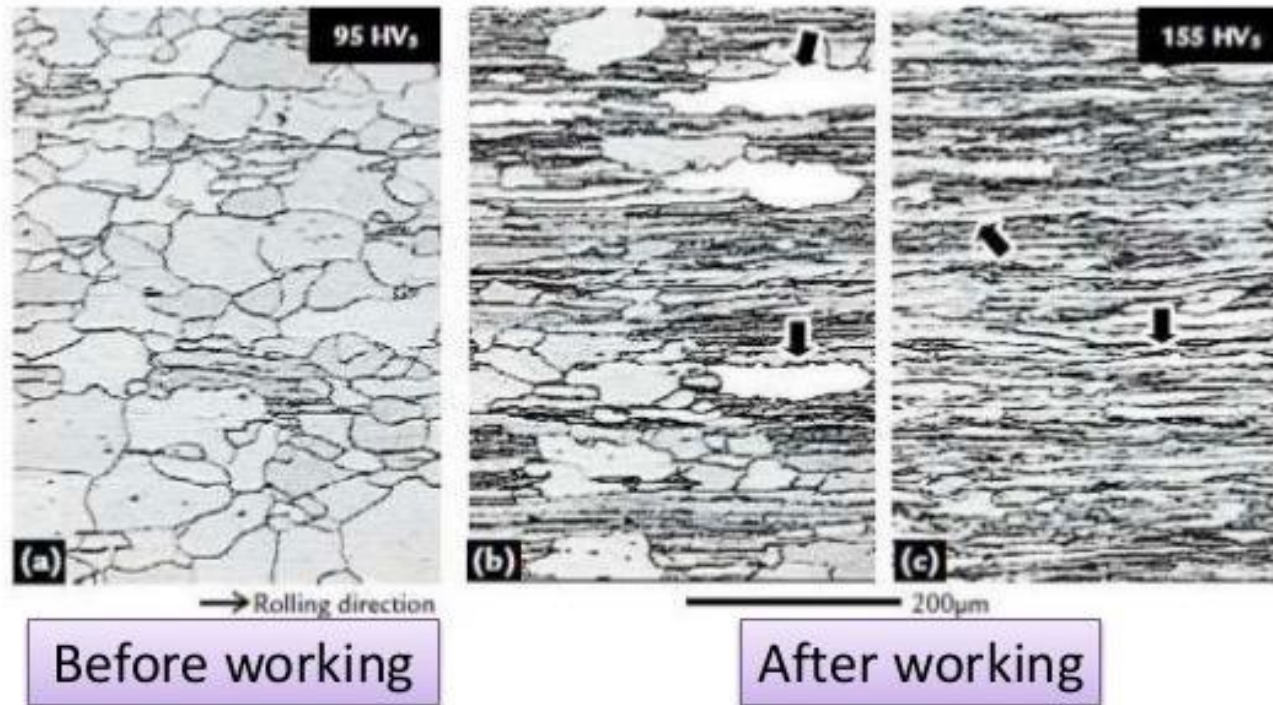
Dislocation motion causes plastic deformation, dislocation pileup causes hardening



https://www.youtube.com/watch?v=JjWdEj_LjZo

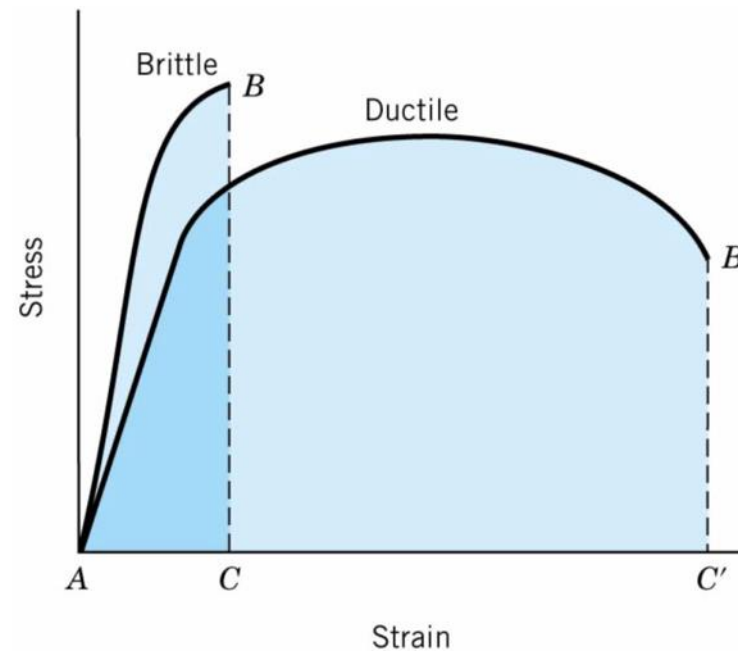
Dislocation Pile Up

- Dislocation motion can be inhibited by barriers, including grain boundaries, precipitates, voids, bubbles, etc.



Ductility

- Ductile materials plastically deform significantly, brittle materials do not
- Quantities defining ductility are total percent elongation at fracture (%EL) and the percent reduction in area (%RA)
- The ductile–brittle transition temperature (DBTT) of a metal is the temperature at which the fracture energy passes below a predetermined value
- Below the DBTT, failure is brittle
- Cold working and neutron irradiation can increase the DBTT, potentially reducing ductility of reactor components



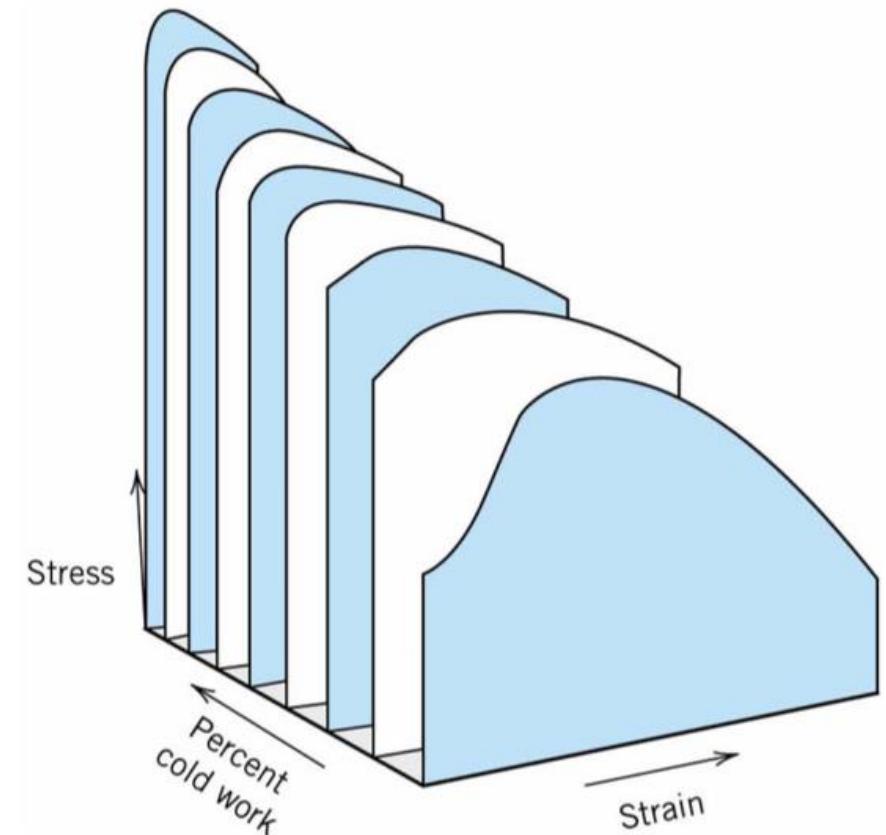
$$\%EL = \frac{(l_f - l_0)}{l_0} \times 100$$

$$\%RA = \frac{(A_0 - A_f)}{A_0} \times 100$$

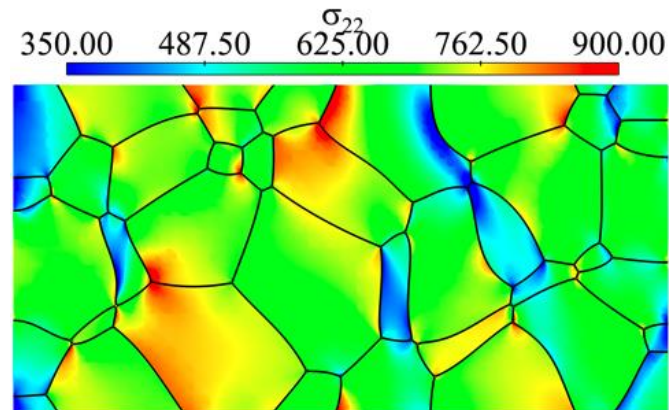
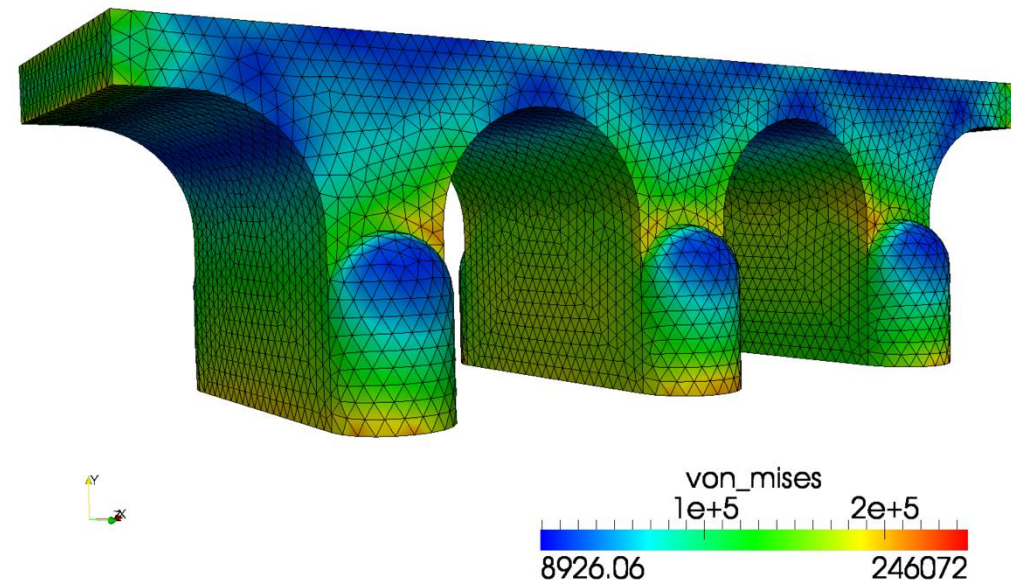
l_f = length at fracture
 A_f = section area at fracture

As the strength increases due to dislocation pile-up, the ductility decreases

- Toughness is the ability of a material to absorb energy and plastically deform without fracturing
- Toughness is related to the area under the stress–strain curve
- In order to be tough, a material must be both strong and ductile
- Materials are often work hardened prior to utilization to modify mechanical properties

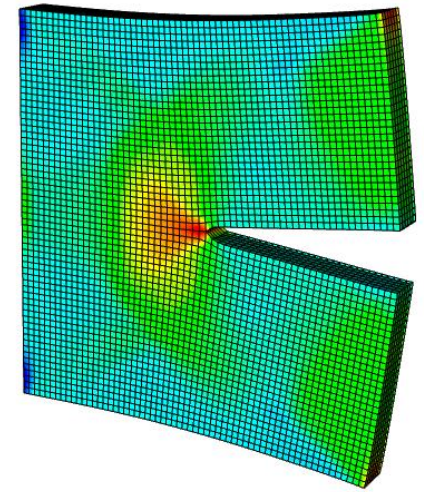
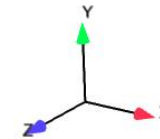


Determine the stress and strain throughout a body



Stress YY (MPa)

438.00
290.05
142.10
-5.85
-153.80



The stress divergence equation derivation

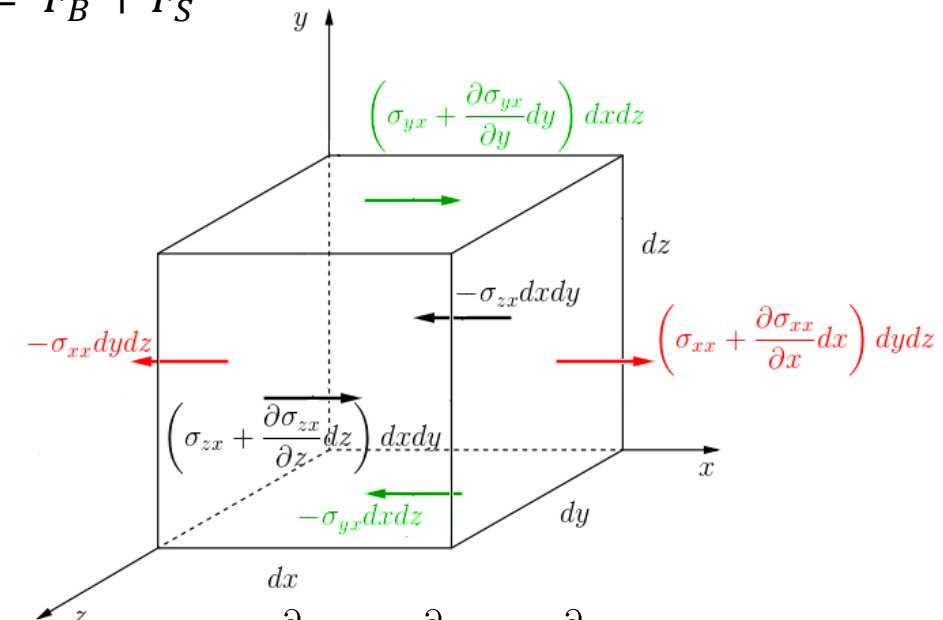
- Generalized momentum conservation: $\frac{d\vec{p}}{dt} = \vec{F} = \vec{F}_B + \vec{F}_S$
- Considering a cubic element, surface forces act on the walls of the cube
- Force on a wall is the product of the stress and the surface area
 - For wall at dx , approximate stress via Taylor expansion

$$\sigma_{xx}(x + dx) = \sigma_{xx}(x) + dx \frac{\partial \sigma_{xx}}{\partial x}$$

$$F_p^x = \left(\sigma_{xx} + \frac{\partial \sigma_{xx}}{\partial x} dx\right) dydz - \sigma_{xx} dydz + \left(\sigma_{yx} + \frac{\partial \sigma_{yx}}{\partial y} dy\right) dx dz - \sigma_{yx} dx dz + \left(\sigma_{zx} + \frac{\partial \sigma_{zx}}{\partial z} dz\right) dx dy - \sigma_{zx} dx dy$$

$$F_p^x = \frac{\partial \sigma_{xx}}{\partial x} dx dy dz + \frac{\partial \sigma_{yx}}{\partial y} dy dx dz + \frac{\partial \sigma_{zx}}{\partial z} dz dx dy$$

$$0 = \nabla \cdot \sigma$$



$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} = 0$$

$$\frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} = 0$$

$$\frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} = 0$$

Simplified Cauchy for our typical system

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}$$

- Assumption 1: We have a static body

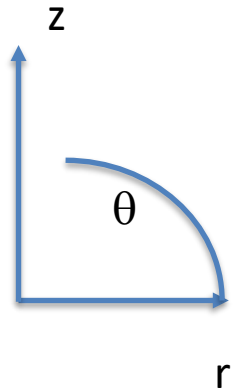
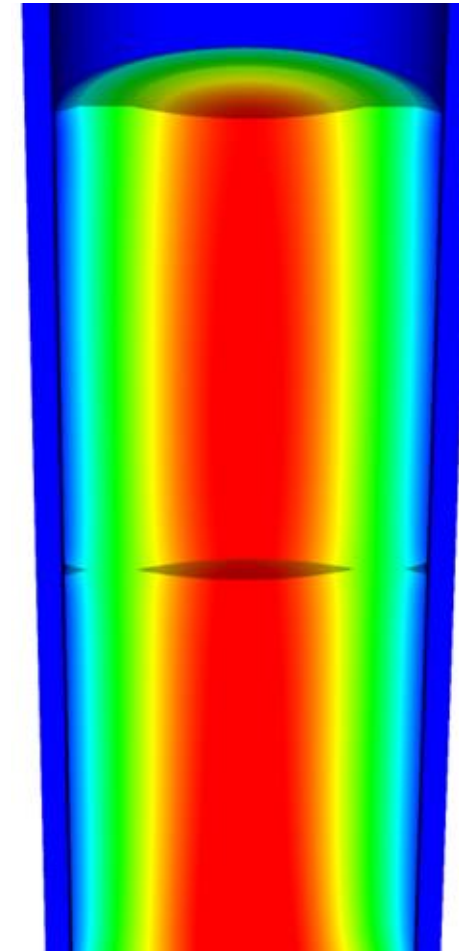
$$0 = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}$$

- Assumption 2: Gravity is negligible

- Assumption 3: The problem is axisymmetric

$$\frac{\partial \sigma_{rr}}{\partial r} + \frac{\sigma_{rr} - \sigma_{\theta\theta}}{r} + \frac{\partial \sigma_{rz}}{\partial z} = 0$$

$$\frac{1}{r} \frac{\partial (r \sigma_{rz})}{\partial r} + \frac{\partial \sigma_{zz}}{\partial z} = 0$$



Solve for the stress throughout a pressurized cylinder (cladding tube) with thin walls

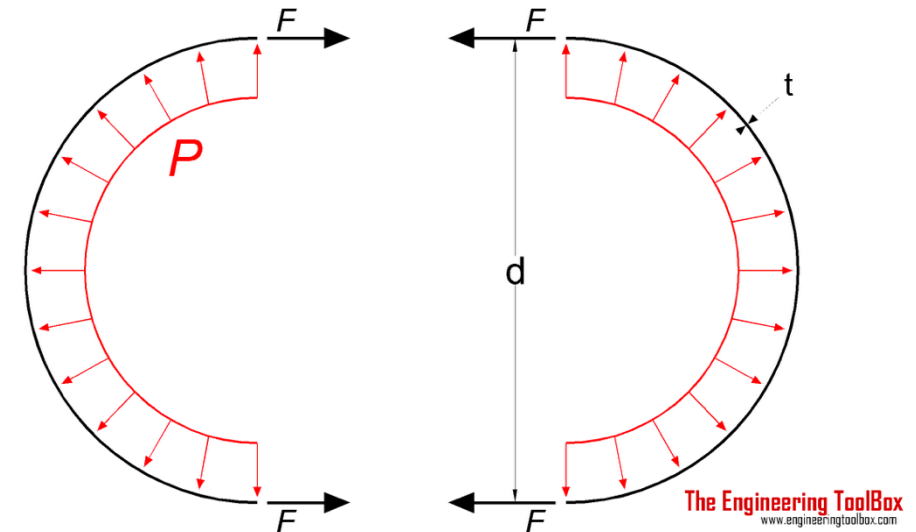
- Because our cylinder has such thin walls (δ = thickness), we can simplify everything to simple force balance
- The hoop stress is the force exerted circumferentially in both directions on every particle in the cylinder wall
- First, we need the Force per unit length due to the pressure

$$F_{\text{press}} = pR \int_0^\pi \sin \theta \, d\theta$$

- Utilize force to hoop stress relation:
- Then we equate the forces and solve for the hoop stress

$$F_{\text{stress}} = 2\delta \, \overline{\sigma_\theta}$$

$$\overline{\sigma_\theta} = \frac{pR}{\delta}$$



Other two stresses for a thin-walled closed cylinder

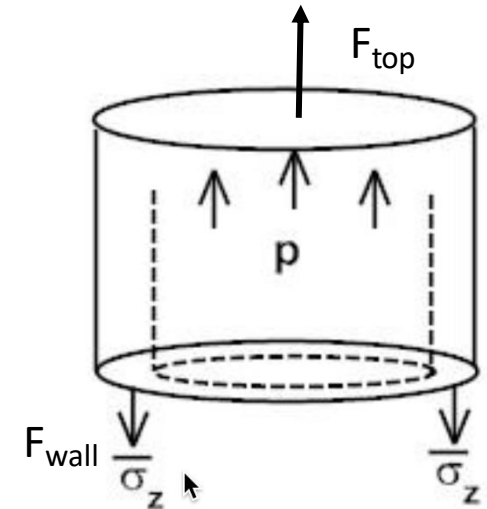
- To find the stress in the z-direction we do another force balance

$$F_{top} = p \pi R^2 \quad F_{wall} = 2\pi R \delta \bar{\sigma}_z$$

- Again, we equate the forces and solve for the stress

$$\bar{\sigma}_z = \frac{pR}{2\delta}$$

- Last, the stress on the inside of the wall is $-p$ and on the outside is zero



$$\bar{\sigma}_r = -\frac{1}{2}p$$

$$\bar{\sigma}_\theta = \frac{pR}{\delta} \quad \bar{\sigma}_z = \frac{pR}{2\delta} \quad \bar{\sigma}_r = -\frac{1}{2}p$$

Determine if a fuel rod will exceed its yield stress

- A fuel rod has an initial pressure of 5 MPa that increases to 9 MPa during operation. Will it exceed its yield stress?
 - $R = 0.55 \text{ cm}$, $\delta = 0.05 \text{ cm}$, $\sigma_y = 381 \text{ MPa}$

$$\bar{\sigma}_\theta = \frac{pR}{\delta} \quad \bar{\sigma}_z = \frac{pR}{2\delta} \quad \bar{\sigma}_r = -\frac{1}{2}p$$

- The largest stress will be the hoop stress
- The hoop stress is $P \cdot (0.55 / .05)$
 - For 5 MPa, $\sigma_\theta = 55 \text{ MPa}$
 - For 9 MPa, $\sigma_\theta = 99 \text{ MPa}$
- With these pressures, we don't come even close to the yield stress of the cladding

Small strain theory

- In continuum mechanics, the infinitesimal strain theory is a mathematical approach to the description of the deformation of a solid body in which the displacements of the material particles are assumed to be much smaller than any relevant dimension of the body so that its geometry and the constitutive properties of the material at each point of space can be assumed to be unchanged by the deformation
- With this assumption, the equations of continuum mechanics are considerably simplified
- The small strain theory is commonly adopted in civil and mechanical engineering for the stress analysis of structures built from relatively stiff elastic materials